

Suggested Teaching Guidelines for
Algorithms and Data Structures using Java
PGCP-AC, August 2026

Duration: 90 Hours: 45 Theory (T) + 45 Lab (L)

Objective: To equip students with problem-solving and computational thinking skills by applying fundamental and advanced data structures, algorithms, and algorithm design techniques to develop efficient and scalable software solutions.

Prerequisites: Knowledge of Java Programming

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment – 20%

List of Course Material

Text book:

1. Object-Oriented Programming and Data Structures using Java, 2026 edition, Cay S. Horstmann / Wiley

Reference book:

1. Data Structures and Algorithms in Java, 6th edition, Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, Subhasish Banerjee / Wiley
2. Algorithms, 4th Edition, Robert Sedgewick, Kevin Wayne / O'Reilly

Sessions 1, 2 & 3: Algorithms & Data Structures (6T + 6L)

- Introductory to Problem Solving & Computational Thinking
- Algorithm Constructs
- Complexity analysis of algorithms (Big O notation)
- OO design: Abstract Data Types (ADTs)
- Basic Data Structures
 - Arrays
 - Stacks
 - Queues
 - Circular Queues

Lab:

- Implement stack through array
- Complexity analysis of loops and recursive algorithms.
- Implement queues with inserting element at different location (First, Last)
- Implement circular queue

Sessions 4, 5 & 6: Linked List (6T + 6L)

- Linked lists
 - Singly linked lists
 - Doubly linked lists
 - Circular linked lists

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- Node-based storage with arrays

Lab:

- Implement circular queue using linked list
- Implement stack using linked list

Sessions 7 & 8: Recursion (4T + 4L)

- Recursion Overview
- Base condition in recursion.
- Direct and indirect recursion.
- Memory allocation to different function calls in recursion.
- Pro and cons of recursion
- Function complexity during recursion

Sessions 9, 10 & 11: Trees & Applications (6T + 6L)

- Introduction to trees
- Trees and terminology
- Tree traversals
- Binary trees
- Complete binary trees / Almost complete binary tree (ACBT)
- Array implementation of ACBT
- Binary search trees
- AVL tree

Lab:

- Write a program to implement a binary search tree and the following operations:
 - Create()
 - Tree traversals: BFS, DFS, PostOrder, InOrder, PreOrder
 - Delete()

Sessions 12, 13 & 14: Searching and Sorting Algorithms (6T + 6L)

- Objectives of Searching
 - The Sequential Search
 - Analysis of Sequential Search
 - The Binary Search
- Analysis of Binary Search
- Introduction to sorting
 - Selection sort
 - Insertion sort
 - Bubble sort
 - Heap sort

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- Merge sort
- Quick sort
- Analysis of sorting algorithms

Lab:

- Writing program to search an item through sequential search technique.
- Implement to find an item in a list through binary search
- Implement sorting algorithm for: insertion sort, Quicksort

Sessions 15 & 16: Hash Functions and Hash Tables (4T + 4L)

- Hashing & Introduction to hash tables
- Hash Functions
- Different type of hash functions
- Collision Resolution
- Linear Probing
- Quadratic Probing
- Double Hashing
- Inserting and deleting an element from a hash table

Lab:

- Implement hashing techniques in different programs solved earlier
- Write a program to implement Hash table
- Fibonacci recursive algorithm improvement using hash table

Sessions 17, 18 & 19: Graphs and Applications (6T + 6L)

- Introduction to Graphs and Graph Terminologies
- Types of Graphs
- Representation of Graphs
 - Adjacency Matrix
 - Adjacency List
 - Graph Traversal Algorithms (Breadth First Search, Depth First Search)
- Shortest Path
 - Level Setting: Dijkstra's algorithm
 - Level Correcting: All-pairs shortest path, Floyd-Warshall algorithm
- Spanning Trees
 - Minimum spanning tree algorithms,
 - Prim's algorithm
 - Kruskal's Algorithm

Lab:

- Implement a graph using adjacency Matrix and traverse using Depth First Search.
- Implement a graph and do traversal using stack and queue.

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Sessions 20, 21 & 22: Algorithm Designs (7T + 7L)

- Writing efficient Algorithms
- Algorithm Design techniques
- Analysis of an Algorithm
 - Asymptotic Analysis
 - Algorithm Analysis
- Analysis of different types of Algorithms
 - Divide and Conquer Algorithm
 - Greedy algorithm
 - Dynamic Programming algorithm
 - Brute force algorithm
 - Backtracking algorithms
 - Branch-and-bound algorithms
 - Stochastic algorithms
- Complexity
 - Complexity Analysis
 - Space complexity of algorithm
 - Time complexity of algorithm
- Case study on Algorithm Design techniques
- Real world Application of Data Structures

Lab:

- Compare different Algorithms previously programmed and do the analysis